

A non-stationary model for statistical arbitrage trading

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Abstract

In this paper we present a non-stationary model for statistical arbitrage trading. This model represents the security price as the sum of an arithmetic Brownian motion and an Ornstein-Uhlenbeck process. A continuous time trading strategy is derived for the process and expressions for the expected value and the variance of the return are formulated. Analytic solutions to problem are obtained in the case where the non-stationary component has zero drift.

Key words: Econophysics, Stochastic Processes, Analytic Solutions

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1 Introduction

Statistical based trading strategies have become widely used by the financial industry over the past decade [1–8]. Such strategies seek to profit by identifying and exploiting predictable behaviour in financial data, for example, short term dependencies in security prices due to inefficiencies such as liquidity. Since the returns on individual trades are usually small, the benefit of a systematic approach lies in being able to perform hundreds or thousands of trades per day, gaining profitability from executing the trade many times. Such an approach incurs a high level of trading cost and can only be implemented by investment banks and specialist market-makers whose transaction costs are virtually negligible. In the context of formulating optimal trading strategies it is essential to model real world data with a stochastic process that reflects the appropriate physical properties of the data. A significant part of designing

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and implementing a strategy involves the use of phenomenological models for price dynamics to model the observed behaviour of the market.

A common approach to statistical based trading attempts to profit from so called mean-reversion trading where traders aim to profit from temporary extremes in pricing, entering a trade when the security reaches an extreme value and exiting when it reverts to some equilibrium level. In this approach, participants form linear combinations of stocks, futures, options and other derivatives in order to produce a synthetic asset whose price follows a mean-reverting stochastic process. The ability to identify the optimal trading points is central to employing such a strategy successfully. There have been several studies of such trading strategies in recent years [1,3,5,8]. Many of these studies model the log-price of the traded securities as stationary processes, such as the Ornstein-Uhlenbeck process. This assumption allows the use of common time invariant statistical measures to estimate trade entry and exit points. However, when the data exhibits non-stationary effects the entry and exit points cannot be reliably determined using methods and models based on stationary time series.

Given a stationary time series it is relatively straightforward to determine whether a given observation represents an extreme value. For data that displays non-stationary behaviour, the task of identifying extreme events becomes more difficult since one must distinguish transient extreme moves from non-stationary shifts in the data. Non-stationary behaviour is a widely observed phenomenon in financial time series [9–19]. Methods such as cointegration [20] have been proposed in order to remove non-stationary effects from time series data, so that analysis may be performed using common statistics based on assumptions of stationarity. In the context of formulating trading strategies, synthetic assets are constructed to remove non-stationary behaviour from price series. In cases such as *dual-listed* securities, where the same stock is traded on multiple exchanges, or *index arbitrage*, where a basket of stocks is traded against an index futures contract, the removal of non-stationary behaviour is possible due to the existence of a deterministic relationship between the assets being traded. However, in more general cases such as the popular method of *pairs trading*, when two similar securities are traded in offsetting positions, it is not possible to guarantee the complete removal of all non-stationary behaviour. If a synthetic asset is constructed when a deterministic relationship does not exist between the component assets, the resulting asset may still retain residual non-stationary behaviour. The magnitude of the remaining non-stationary behaviour may have a significant effect on the return and risk of the strategy.

In this paper we present a method for the construction of optimal trading strategies on non-stationary securities. We consider the price of the security to be a combination of stationary and non-stationary components. We propose

a model for asset dynamics as an Ornstein-Uhlenbeck process with stochastic mean. The model exhibits mean-reverting behaviour around an arithmetic Brownian motion. The optimal trading strategy is calculated for the chosen stochastic process under the framework described in [7,8]. Under this model, the expected trade length is independent of the non-stationary behaviour. Expressions for the expected profit rate and its variance are presented. In this case the expected profit rate is shown to be a function of the drift of the non-stationary component, while the variance of the profit rate is shown to be a function of the instantaneous variance of the non-stationary component. We derive analytic solutions for the expected return and the variance of the return for the strategy in the case where the non-stationary component has zero drift. The expected return is shown to be independent of the non-stationary behaviour while the variance is shown to be a linear function of the instantaneous variance of the non-stationary component. We derive analytic solutions for the strategy that maximises the expected return. We derive a closed form solution for the variance associated with this optimal case. The results for the mean and variance of the strategy are used to derive an expression for the signal-to-noise ratio of the strategy. This ratio is used to determine an optimal strategy in terms of the strength of the non-stationary component. The model is illustrated with a real-world example where the stationary and non-stationary components are directly observable. The results for the optimal strategies are calculated for this example.

The rest of the paper is as follows. In the following section we present the model and define the continuous trading strategy for the process. In Section 3 we derive analytic solutions for the expected return and the variance of the return. An analytic solution for the problem of maximising the expected return is also presented. Section 4 presents the results of applying the model to real world data. Section 5 summarises the main results and concludes.

2 Non-stationary model for trading

In order to address the potential existence of non-stationary behaviour we assume that a synthetic asset consists of two components: a stationary process that provides transient mean-reversion effects over short time scales; and non-stationary process that drives the long term behaviour of the asset. We express the price of the synthetic security p_t as,

$$p_t = e^{S_t},$$

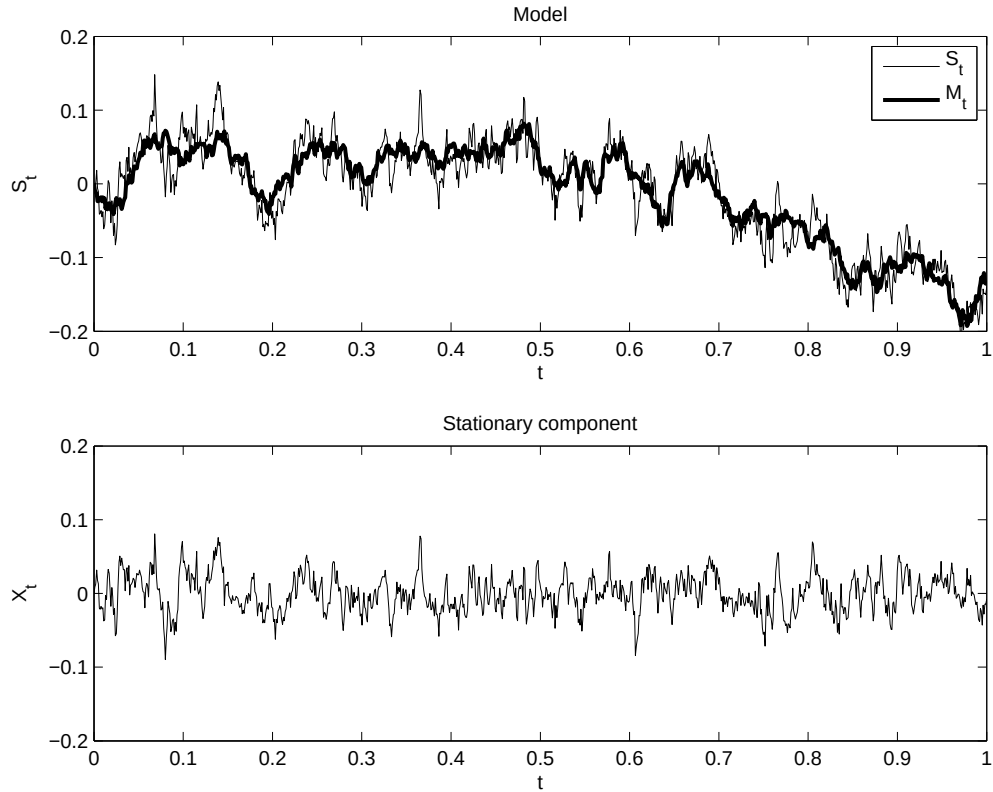


Fig. 1. A sample path for the model showing the non-stationary and stationary components. This example uses parameters $\alpha = 200$, $\eta = 0.5$, $\gamma = 0$, and $\theta = 0.2$ with $\Delta t = 1/250$.

and propose modelling the observed log-price, S_t , as a superposition of stochastic processes,

$$S_t = X_t + M_t, \quad (1)$$

where X_t and M_t satisfy the stochastic differential equations,

$$dX_t = -\alpha X_t dt + \eta dW_t, \quad (2)$$

$$dM_t = \gamma dt + \theta dZ_t. \quad (3)$$

Here Z_t and W_t are independent Wiener processes,

$$\text{corr}(W_t, Z_t) = 0.$$

A simulated sample path of the process S_t is displayed in Figure 1. In this figure we can see that the process displays mean-reversion behaviour around a non-stationary random walk. While the process S_t is non-stationary, it displays local stationary behaviour on small time scales. A strategy that attempts

to profit from this localised stationary behaviour can be constructed by identifying extreme deviation of the process X_t .

We consider a trading strategy that enters a trade by either buying the asset when it is cheap, or by selling the asset when it is expensive. Since we are seeking to exploit the stationary component of S_t , the trade entry and exit points, b_1, b_2 with $b_1 < b_2$ are expressed in terms of X_t . The continuous time trading strategy we define for this process consists of two types of trades. The first type corresponds to buying the security and is defined by entering a trade when $X_t = b_1$, exiting the trade at $X_t = b_2$, where $b_1 < b_2$. The second type corresponds to selling the security and is defined by entering when $X_t = b_2$, exiting the trade when $X_t = b_1$. When these two types are combined, the resulting sequence of trades completes a trading cycle that starts and finishes at the value $X_t = b_1$. Such a strategy can be thought of as periodic, since the actions are repeated between trade entry points. However, since X_t is a stochastic process, the time taken to complete the trade cycle will be a random variable $\mathcal{T}$. We refer to $\mathcal{T}$ as the total trade length. Thus, the behaviour of $\mathcal{T}$ will largely determine the properties of the strategy.

We decompose the total trade length into sub intervals,

$$\mathcal{T} = \mathcal{T}_1 + \mathcal{T}_2,$$

where $\mathcal{T}_1$ is the time taken for the process X_t to travel from b_1 to b_2 , completing a buy-trade, and $\mathcal{T}_2$ is the time taken from b_2 back to b_1 , completing a sell-trade. Here the variables $\mathcal{T}_1, \mathcal{T}_2$ can be identified as first-passage times for the process X_t . Furthermore, since X_t is a Markov process, the passage times $\mathcal{T}_1, \mathcal{T}_2$ are independent.

Expressions for the return per unit time and variance of the return per unit time of the strategy, can be formulated in terms of the trading frequency. Let $r(b_1, b_2, c)$ be the return for the trading cycle as a function of entry and exit levels and transaction cost, c . Then, the expected value and the variance of the return for the strategy are given by,

$$\mu(b_1, b_2, c) = \mathbb{E}[r(b_1, b_2, c)/\mathcal{T}], \quad (4)$$

$$\sigma^2(b_1, b_2, c) = \mathbb{V}[r(b_1, b_2, c)/\mathcal{T}]. \quad (5)$$

The signal to trade is generated by the stationary process X_t , however the return for the trading cycle is a function of S_t . This means that in this case, unlike the Ornstein-Uhlenbeck process [8], the return over the trade cycle will be stochastic. To calculate the return for the trading cycle we consider the return over $\mathcal{T}_1$ and the return over $\mathcal{T}_2$,

$$\begin{aligned}
r(b_1, b_2, c) &= (S_{t+\mathcal{T}_1} - S_t) - (S_{t+\mathcal{T}_1+\mathcal{T}_2} - S_{t+\mathcal{T}_1}), \\
&\stackrel{d}{=} 2(b_2 - b_1 - c) + M_{\mathcal{T}_1} - M_{\mathcal{T}_2}, \\
&\stackrel{d}{=} 2(b_2 - b_1 - c) + \gamma(\mathcal{T}_1 - \mathcal{T}_2) + \theta\sqrt{\mathcal{T}_1 + \mathcal{T}_2}\varepsilon,
\end{aligned} \tag{6}$$

where $\stackrel{d}{=}$ denotes equality in distribution and $\varepsilon \sim \mathcal{N}(0, 1)$. Thus the return over a single trade cycle is given by the difference between the entry and exit points, minus transaction costs, plus any movement in the spread due to the non-stationary effects.

Using Eqs. (4), (5) and (6), the expected return for the strategy can be formulated as,

$$\begin{aligned}
\mu(b_1, b_2, c) &= \mathbb{E} \left[\frac{2(b_2 - b_1 - c) + \gamma(\mathcal{T}_1 - \mathcal{T}_2) + \theta\sqrt{\mathcal{T}_1 + \mathcal{T}_2}\varepsilon}{\mathcal{T}_1 + \mathcal{T}_2} \right], \\
&= \mathbb{E} \left[\frac{2(b_2 - b_1 - c) + \gamma(\mathcal{T}_1 - \mathcal{T}_2)}{\mathcal{T}_1 + \mathcal{T}_2} \right],
\end{aligned}$$

and the variance of the return as,

$$\begin{aligned}
\sigma^2(b_1, b_2, c) &= \mathbb{E} \left[\left(\frac{2(b_2 - b_1 - c) + \gamma(\mathcal{T}_1 - \mathcal{T}_2) + \theta\sqrt{\mathcal{T}_1 + \mathcal{T}_2}\varepsilon}{\mathcal{T}_1 + \mathcal{T}_2} \right)^2 \right] - \mu^2, \\
&= \mathbb{V} \left[\frac{2(b_2 - b_1 - c) + \gamma(\mathcal{T}_1 - \mathcal{T}_2)}{\mathcal{T}_1 + \mathcal{T}_2} \right] + \mathbb{E} \left[\frac{\theta^2}{\mathcal{T}_1 + \mathcal{T}_2} \right].
\end{aligned}$$

It is clear from these equations that the non-stationary component of the model contributes to the expected return only through the drift parameter, γ . However, the variance of the return is affected by both the drift and the instantaneous variance of the non-stationary component. These equations can be solved numerically using either Monte-Carlo simulation or quadrature, after obtaining the distribution of $\mathcal{T}_1$ and $\mathcal{T}_2$ by solving the corresponding Fokker-Planck equations as described in [7]. However, the equations for the expected return and the variance of the return can be treated analytically in the case where the non-stationary component has zero drift, *i.e.* $\gamma = 0$.

3 Analytic solutions for the case $\gamma = 0$

In the case where the non-stationary component M_t has zero drift, the equations for the mean and variance of the strategy return reduce to,

$$\mu(b_1, b_2, c) = 2(b_2 - b_1 - c)\mathbb{E}[1/\mathcal{T}],$$

$$\sigma^2(b_1, b_2, c) = 4(b_2 - b_1 - c)^2 \mathbb{V}[1/\mathcal{T}] + \theta^2 \mathbb{E}[1/\mathcal{T}].$$

Since the value of $\mathcal{T}$ is independent for each trade cycle, the trade frequency is a renewal process and we can make use of the following results from renewal theory [21],

$$\begin{aligned}\mathbb{E}[1/\mathcal{T}] &\sim 1/\mathbb{E}[\mathcal{T}], \\ \mathbb{V}[1/\mathcal{T}] &\sim \mathbb{V}[\mathcal{T}]/\mathbb{E}[\mathcal{T}]^3,\end{aligned}$$

to obtain

$$\mu(b_1, b_2, c) = 2(b_2 - b_1 - c)/\mathbb{E}[\mathcal{T}], \quad (7)$$

$$\sigma^2(b_1, b_2, c) = 4(b_2 - b_1 - c)^2 \mathbb{V}[\mathcal{T}]/\mathbb{E}[\mathcal{T}]^3 + \theta^2/\mathbb{E}[\mathcal{T}]. \quad (8)$$

Closed form solutions for the expected value and the variance of the length of the trade cycle $\mathcal{T}$ are given by [8] as,

$$\mathbb{E}[\mathcal{T}] = \frac{\pi}{\alpha} \left(\text{Erfi} \left(b_2 \sqrt{\alpha}/\eta \right) - \text{Erfi} \left(b_1 \sqrt{\alpha}/\eta \right) \right), \quad (9)$$

$$\mathbb{V}[\mathcal{T}] = \frac{\left(w_1 \left(\frac{b_2 \sqrt{2\alpha}}{\eta} \right) - w_1 \left(\frac{b_1 \sqrt{2\alpha}}{\eta} \right) - w_2 \left(\frac{b_2 \sqrt{2\alpha}}{\eta} \right) + w_2 \left(\frac{b_1 \sqrt{2\alpha}}{\eta} \right) \right)}{\alpha^2}, \quad (10)$$

where

$$\begin{aligned}w_1(z) &= \left(\frac{1}{2} \sum_{k=1}^{\infty} \Gamma(k/2) \left(\sqrt{2}z \right)^k / k! \right)^2 - \left(\frac{1}{2} \sum_{k=1}^{\infty} (-1)^k \Gamma(k/2) \left(\sqrt{2}z \right)^k / k! \right)^2, \\ w_2(z) &= \sum_{k=1}^{\infty} \Gamma((2k-1)/2) \Psi((2k-1)/2) \left(\sqrt{2}z \right)^{(2k-1)} / (2k-1)!,\end{aligned}$$

and $\Psi(x) = \psi(x) - \psi(1)$ and $\psi(x)$ is the digamma function,

$$\psi(z) = \frac{1}{\Gamma(z)} \frac{d}{dz} \Gamma(z).$$

Using Eqs. (7), and (9) we can obtain the analytic form of the expected return for the strategy,

$$\mu(b_1, b_2, c) = \frac{2\alpha(b_2 - b_1 - c)}{\pi \left(\text{Erfi} \left(\frac{b_2 \sqrt{\alpha}}{\eta} \right) - \text{Erfi} \left(\frac{b_1 \sqrt{\alpha}}{\eta} \right) \right)}. \quad (11)$$

Likewise from Eqs. (8), (10) and (9), the variance of the strategy return is,

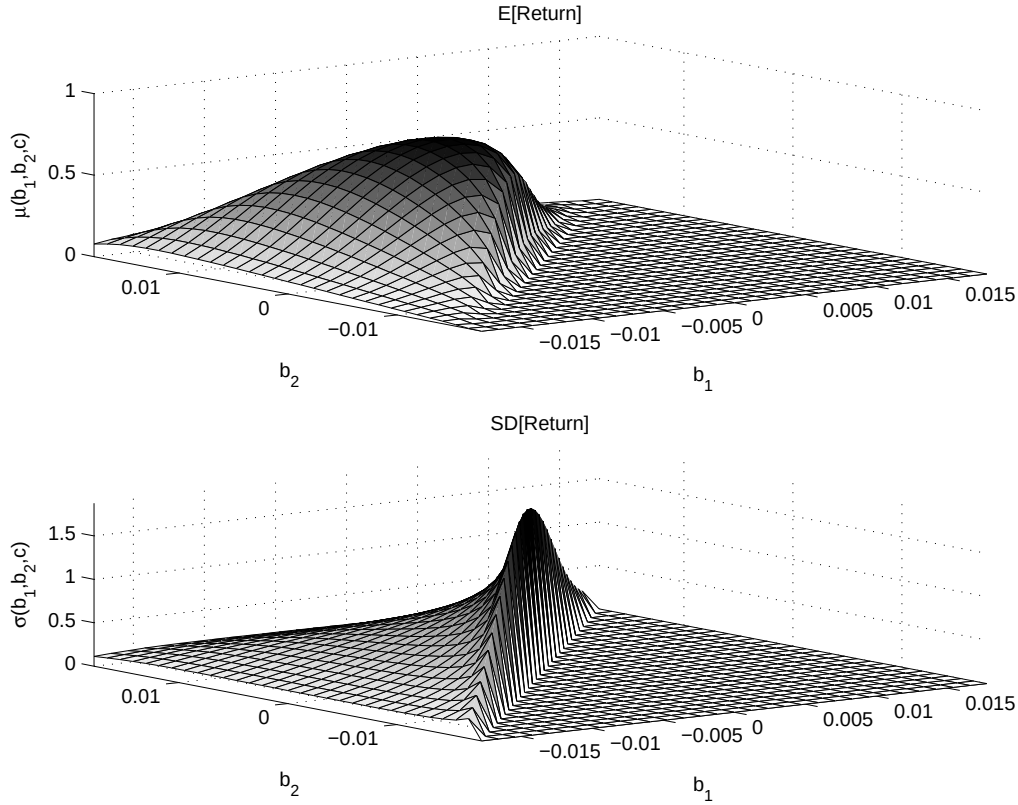


Fig. 2. Surfaces plots for the expected return and the variance of the return in the region $b_2 - b_1 \geq c$ for the case of zero drift. The variance surface has a singularity at $b_1 = b_2$. This example uses parameter values $\alpha = 199.9159$, $\eta = 0.1185$, $\gamma = 0$, $\theta = 0.0943$ and $c = 0.001$.

$$\begin{aligned} \sigma^2(b_1, b_2, c) &= 4\alpha(b_2 - b_1 - c)^2 \\ &\times \frac{\left(w_1 \left(\frac{b_2\sqrt{2\alpha}}{\eta}\right) - w_1 \left(\frac{b_1\sqrt{2\alpha}}{\eta}\right) - w_2 \left(\frac{b_2\sqrt{2\alpha}}{\eta}\right) + w_2 \left(\frac{b_1\sqrt{2\alpha}}{\eta}\right)\right)}{\pi^3 \left(\operatorname{Erfi} \left(\frac{b_2\sqrt{\alpha}}{\eta}\right) - \operatorname{Erfi} \left(\frac{b_1\sqrt{\alpha}}{\eta}\right)\right)^3} \\ &+ \frac{\alpha\theta^2}{\pi \left(\operatorname{Erfi} \left(\frac{b_2\sqrt{\alpha}}{\eta}\right) - \operatorname{Erfi} \left(\frac{b_1\sqrt{\alpha}}{\eta}\right)\right)}. \end{aligned} \quad (12)$$

These analytic formulae, for the mean and variance, determine the properties of the strategy in terms of the trade entry and exit levels and transaction costs. The results obtained above are almost identical to those obtained for the Ornstein-Uhlenbeck process by [8], the only significant difference being an additive term in the equation for the variance of the return. Figure 2 displays surface plots of the expected value and the variance of the return in the region $b_2 - b_1 \geq c$. This region does not include the singularity in the return variance which occurs at $b_1 = b_2$ and therefore is not displayed in the figure.

As in [8] the optimal strategy that maximises the expected return occurs when $b_2 = -b_1$ with $b_1 < 0$. Accordingly the maximum expected return is given by,

$$\mu^*(b_1, c) = \frac{\alpha(2b_1 + c)}{\pi \operatorname{Erfi}(b_1 \sqrt{\alpha}/\eta)}, \quad (13)$$

where the optimal value of a satisfies the following equation,

$$e^{\frac{\alpha b_1^2}{\eta^2}}(2b_1 + c) - \eta \sqrt{\frac{\pi}{4\alpha}} \operatorname{Erfi}(b_1 \sqrt{\alpha}/\eta) = 0.$$

Further, using the identities, $w_1(z) = -w_1(-z)$, $w_2(z) = -w_2(-z)$ the corresponding variance of the return in the optimal case is given by,

$$\sigma^2(b_1, c) = \alpha(2b_1 + c)^2 \frac{\left(w_1\left(\frac{b_1 \sqrt{2\alpha}}{\eta}\right) - w_2\left(\frac{b_1 \sqrt{2\alpha}}{\eta}\right) \right)}{\pi^3 \operatorname{Erfi}\left(\frac{b_1 \sqrt{\alpha}}{\eta}\right)^3} - \frac{\alpha \theta^2}{2\pi \operatorname{Erfi}\left(\frac{b_1 \sqrt{\alpha}}{\eta}\right)}. \quad (14)$$

Note that maximising the expected return does not take into account the behaviour of the non-stationary component of the model. Equations (12) and (14) show that the variance of the return is a linear function of the variance of the non-stationary term, θ^2 . By considering the variance of the return as a measure of risk, it is clear that as θ^2 increases, the non-stationary component tends to dominate the risk associated with the strategy. This result indicates that a more suitable choice of objective function should incorporate the variance of the return.

In order to include the variance of the strategy into the problem of determining an optimal strategy, we consider an objective function that measures the magnitude of the expected return with respect to the variance. Accordingly, we define the objective function as the following signal-to-noise ratio,

$$\begin{aligned} S(b_1, b_2, c) &= \sqrt{\mu^2(b_1, b_2, c)/\sigma^2(b_1, b_2, c)}, \\ &= \sqrt{\frac{4(b_2 - b_1 - c)^2 \mathbb{E}[\mathcal{T}]}{4(b_2 - b_1 - c)^2 \mathbb{V}[\mathcal{T}] + \theta \mathbb{E}[\mathcal{T}]^2}}. \end{aligned} \quad (15)$$

Since we are only considering the region $b_2 - b_1 \geq c$, then this measure is analogous to the Sharpe ratio [22] with zero risk free rate. The non-stationary behaviour of the model has clear effect in Eq. (15) though the parameter θ , as evidenced by the fact that $S(b_1, b_2, c) \rightarrow 0$ as $\theta \rightarrow \infty$. By considering the expected value and the variance of $\mathcal{T}$ as functions of b_1 and b_2 , we can use the derivatives, $\frac{\partial S}{\partial b_1} = \frac{\partial S}{\partial b_2} = 0$, to obtain the following equations,

$$2(b_1 - b_2 + c)^4 \left(\frac{\theta \mathbb{E}[\mathcal{T}]^2}{(b_1 - b_2 + c)} + 2\mathbb{V}[\mathcal{T}] \frac{\partial}{\partial b_1} \mathbb{E}[\mathcal{T}] - 2\mathbb{E}[\mathcal{T}] \frac{\partial}{\partial b_1} \mathbb{V}[\mathcal{T}] \right) = 0,$$

$$-2(b_1 - b_2 + c)^4 \left(\frac{\theta \mathbb{E}[\mathcal{T}]^2}{(b_1 - b_2 + c)} - 2\mathbb{V}[\mathcal{T}] \frac{\partial}{\partial b_1} \mathbb{E}[\mathcal{T}] + 2\mathbb{E}[\mathcal{T}] \frac{\partial}{\partial b_1} \mathbb{V}[\mathcal{T}] \right) = 0.$$

From the equations for the expected time Eq. (9) and variance Eq. (10) we have,

$$\begin{aligned} \frac{\partial}{\partial b_1} \mathbb{E}[\mathcal{T}] &= -\frac{\partial}{\partial b_2} \mathbb{E}[\mathcal{T}], \\ \frac{\partial}{\partial b_1} \mathbb{V}[\mathcal{T}] &= -\frac{\partial}{\partial b_2} \mathbb{V}[\mathcal{T}]. \end{aligned}$$

Thus, it is clear that we must again have $b_1 < 0$ and $b_2 = -b_1$ as a condition to optimise the strategy. Using Eqs. (15), (9), (10) and noting that $w_1(-z) = -w_1(z)$ and $w_2(-z) = -w_2(z)$, we can write the signal-to-noise ratio as,

$$S^*(b_1, c) = \sqrt{\frac{(2b_1 + c)^2 \alpha \pi \operatorname{Erfi}\left(\frac{b_1 \sqrt{\alpha}}{\eta}\right)}{(2b_1 + c)^2 \left(w_1\left(\frac{b_1 \sqrt{2\alpha}}{\eta}\right) + w_2\left(\frac{b_1 \sqrt{2\alpha}}{\eta}\right)\right) - \frac{1}{2} \theta \pi^2 \operatorname{Erfi}\left(\frac{b_1 \sqrt{\alpha}}{\eta}\right)^2}}. \quad (16)$$

which may be maximised to find b_1 by applying a straightforward optimisation routine.

4 Results

We illustrate the results of the previous section with an example synthetic asset constructed from the dual-listed pair of securities, ANZ.AX and ANZ.NZ. The issued securities are for the same company but are quoted in different currencies on the different exchanges. The non-stationary behaviour is due primarily to changes in the foreign exchange rate and is therefore directly observable. This scenario presented in this example is somewhat superfluous, since the synthetic security can be made stationary by trading the exchange rate as in [7,8]. However, this example provides a straightforward way to illustrate and test the model. The synthetic asset is constructed as a linear combination of the two securities such that the log-price is given by,

$$S_t = \log p_1 - \log p_2.$$

where p_1 , p_2 are the prices of ANZ.AX and ANZ.NZ respectively. Figure 3 displays the log-price, S_t of the synthetic asset and the non-stationary and stationary components, M_t and X_t . This figure clearly shows that the FX rate accounts for all of the non-stationary behaviour of S_t . The residual component,

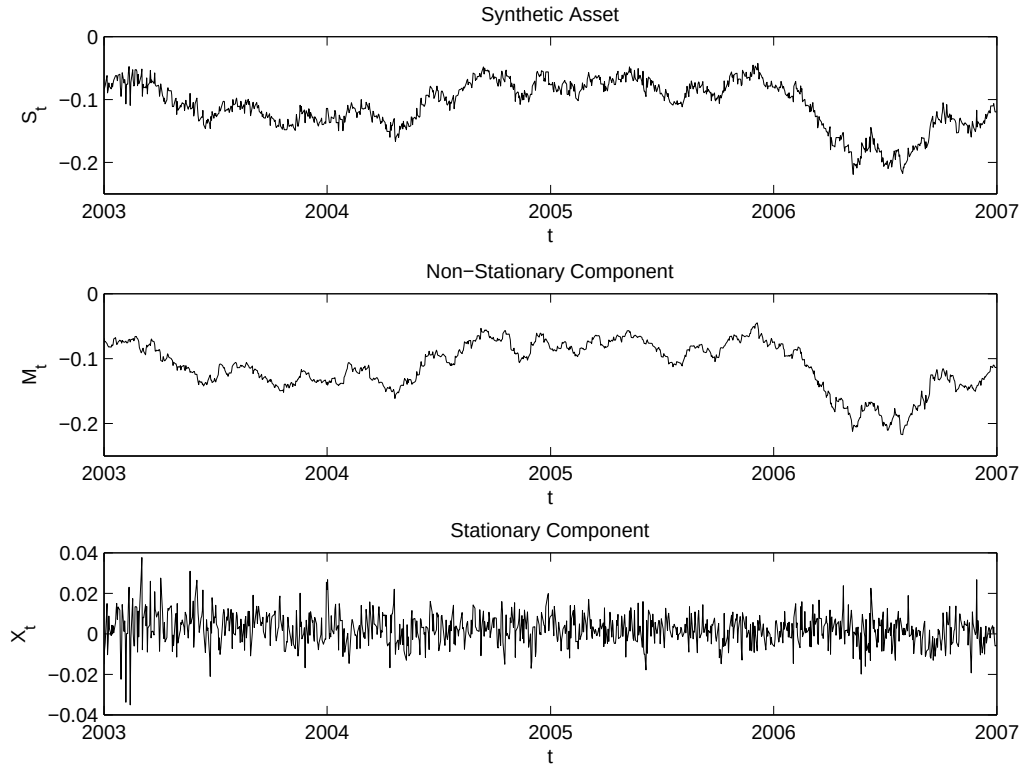


Fig. 3. Example synthetic asset constructed from ANZ dual listed securities. The stationary component is obtained by subtracting the foreign exchange rate log-price.

	α	η	γ	θ
Estimate:	199.9159	0.1185	0.0012	0.0943
Std Error:	7.5118	0.0247	0.0459	0.0197

Table 1

Parameter estimates for the stationary and non-stationary components.

X_t , clearly displays stationary behaviour. By fitting the model to the observed components we can obtain estimates for α , η , γ and θ . Table 1 displays estimated parameter values for the example obtained via maximum likelihood estimation on Eqs. (2) and (3). The standard error for the estimates is also included in the table.

The estimated value of the non-stationary drift parameter $\gamma = 0.0012 \pm 0.0459$ justifies making the approximation, $\gamma \approx 0$. Therefore, for this example, we may consider the strategy for the zero drift case as presented in Section 3. Figure 4 displays the maximum expected return and corresponding variance obtained by solving Eq. (13) and (14). Table 2 presents the corresponding values for different transaction costs.

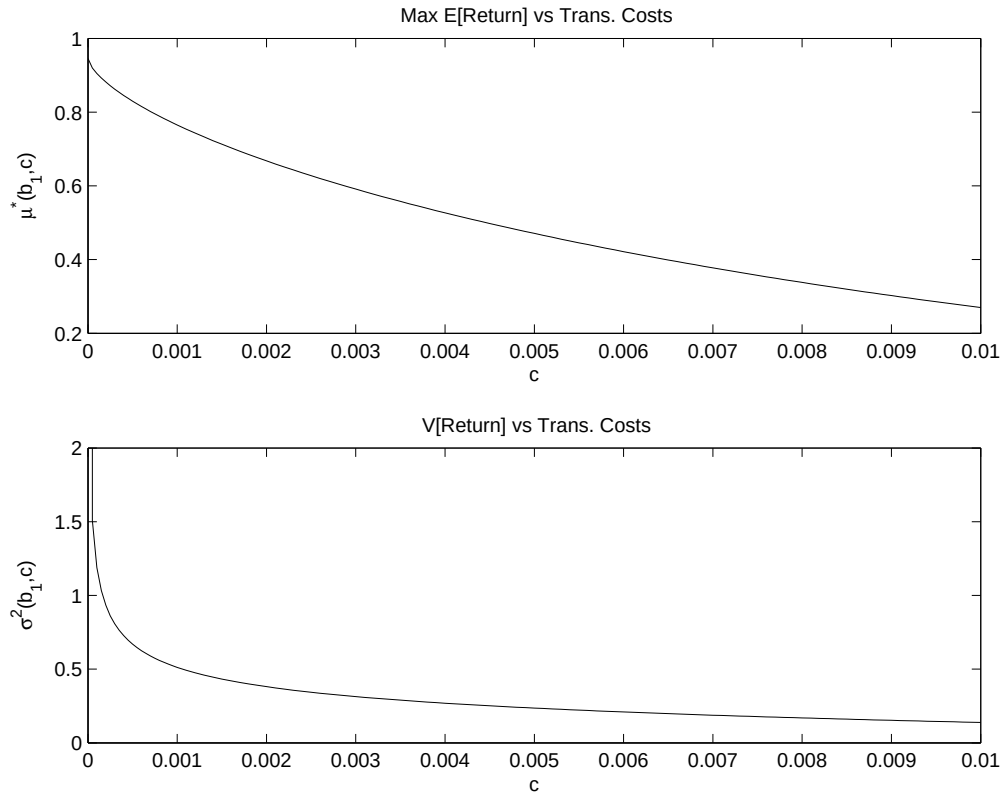


Fig. 4. Optimal expected return and variance as a function of transaction cost. This example uses parameters $\alpha = 199.9159$, $\eta = 0.1185$, $\gamma = 0$ and $\theta = 0.0943$.

c	b_1	$\mu^*(b_1, c)$	$\sigma^2(b_1, c)$
0.001	-0.0038540	0.76512	0.51212
0.002	-0.0049403	0.66782	0.38110
0.003	-0.0057415	0.59122	0.31368
0.004	-0.0064086	0.52678	0.26900
0.005	-0.0069964	0.47089	0.23581
0.006	-0.0075315	0.42155	0.20947
0.007	-0.0080294	0.37753	0.18766
0.008	-0.0084998	0.33796	0.16907
0.009	-0.0089495	0.30224	0.15287
0.010	-0.0093831	0.26990	0.13854

Table 2

Results for maximising the expected return for different values of c with $\alpha = 199.9159$, $\eta = 0.1185$, $\gamma = 0$, $\theta = 0.0943$

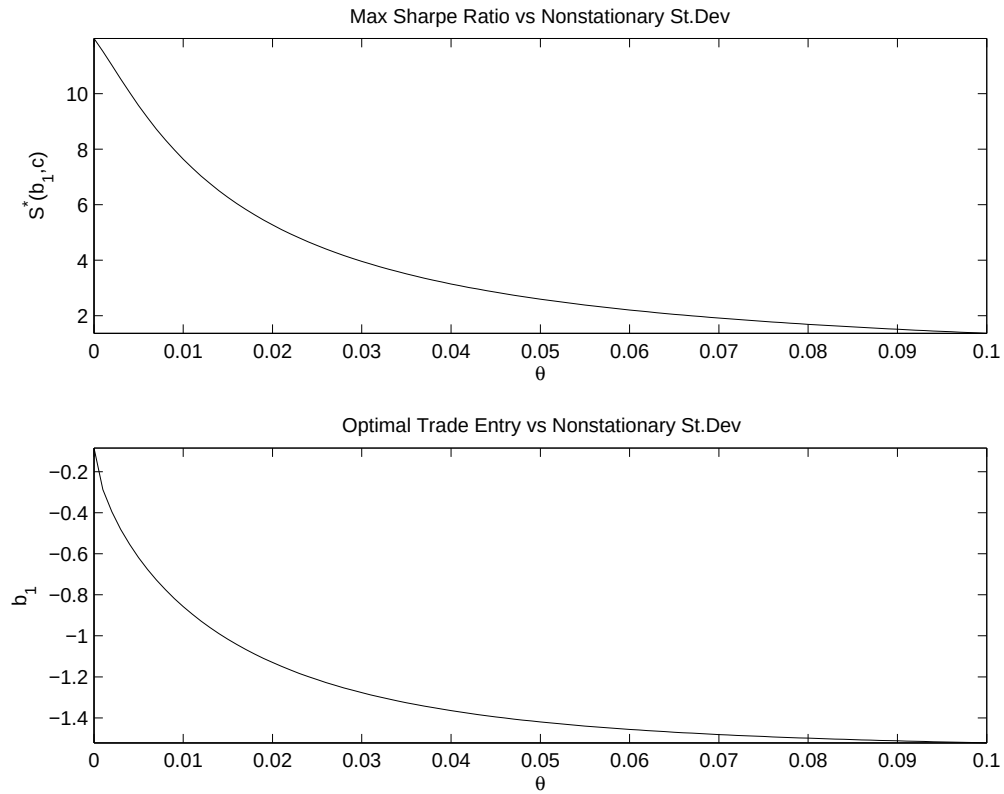


Fig. 5. Optimal values for $S(b_1, c)$ and b_1 as a function of θ . This example uses parameters $\alpha = 199.9159$, $\eta = 0.1185$, $\gamma = 0$, and $c = 0.001$.

θ	b_1	$S^*(b_1, c)$	$\mu(b_1, c)$	$\sigma^2(b_1, c)$
0.01	-0.0050832	7.63855	0.74933	0.00962
0.02	-0.0066953	5.27058	0.69393	0.01734
0.03	-0.0075676	3.95730	0.65226	0.02717
0.04	-0.0080860	3.14332	0.62444	0.03947
0.05	-0.0084128	2.59607	0.60589	0.05447
0.06	-0.0086287	2.20569	0.59325	0.07234
0.07	-0.0087771	1.91451	0.58440	0.09318
0.08	-0.0088825	1.68964	0.57803	0.11703
0.09	-0.0089597	1.51110	0.57333	0.14396
0.10	-0.0090176	1.36610	0.56978	0.17396

Table 3

Results for maximising the Sharpe ratio for different values of θ with $\alpha = 199.9159$, $\eta = 0.1185$, $\gamma = 0$, $c = 0.001$.

	b_1	$\mu(b_1, c)$	$\sigma^2(b_1, c)$	$S(b_1, c)$
$\max \mu$	-0.0038540	0.76512	0.51212	1.0692
$\max S$	-0.0089865	0.57169	0.15648	1.4452

Table 4

Comparison of results for maximising the expected return and the Sharpe ratio for $\alpha = 199.9159$, $\eta = 0.1185$, $\gamma = 0$, $\theta = 0.0943$, $c = 0.001$.

Figure 5 shows the results of maximising the Sharpe ratio for different values of θ . The optimal Sharpe ratio decreases rapidly as the strength of the non-stationary noise is increased through the term θ . Likewise the optimal trading bands widen rapidly with θ . This effect illustrates how the optimal strategy requires a higher return per trade to offset the variance due to the increasing strength of the non-stationary behaviour. Table 4 provides a comparison of the results of maximising the two objective functions using the same data. We can see that the variance of the returns is much higher for the case of maximising the expected return. The optimal trading bands are approximately three times wider in the case of maximising the Sharpe ratio. This suggests that in the presence of non-stationary behaviour, a wider set of bands increases the return per unit risk of the strategy. In this case, the return per trade that is obtained from using wider trading bands tends to offset some of the variance due to non-stationary behaviour. These results are significant in a practical sense since they show how profitability and risk of such a strategy are affected by the strength of any non-stationary behaviour in the traded security. Further, the results illustrate the large difference between optimising different objective functions and highlights the importance that non-stationary behaviour can have in choosing a suitable objective function.

For more general synthetic securities, neither of the processes M_t and X_t are directly observable. In cases where the components are not observable, the model parameters may be estimated by employing methods such as state-space filtering [23].

5 Summary

In this paper we have presented a method for constructing statistical arbitrage trading strategies for assets displaying non-stationary data. This model represents an attempt to explicitly incorporate the effects of non-stationary behaviour on a trading strategy that exploits extreme movements in price over short time scales. We proposed a model for asset price dynamics that is constructed as a linear combination of stationary and non-stationary processes. The model exhibits mean reverting behaviour around a non-stationary ran-

dom walk. The stationary component was modelled as an Ornstein-Uhlenbeck process, while the non-stationary component was modelled by an arithmetic Brownian motion. Non-stationary behaviour is an important feature to consider in the construction of statistical based trading strategies. A continuous time trading strategy was formulated for the model and expressions for the expected return and the variance of the return were formulated. These expressions show that the expected return is a function of the drift of the non-stationary component, while the return variance is a linear function of the instantaneous variance of the non-stationary component. We obtained analytic solutions for expected return and its variance. In the case where the non-stationary component exhibits zero drift. These closed form solutions represent measures for the return and the risk of the strategy. It was shown that in the zero drift case the expected return is independent of the non-stationary term. An analytic solution to the problem of maximising the expected return was presented together with an expression for the variance of the return in this case. An expression for the signal-to-noise ratio of the strategy was derived. This ratio allows for the determination of optimal strategies in terms of the variance of the non-stationary behaviour of the traded asset. The model and strategy were illustrated with a real-world example where the stationary and non-stationary components are observable. In general, these two components will not be observable.

The methods described in this paper, in particular the analytic solutions, provide a way to investigate and understand the behaviour of statistical trading strategies. It is well known that financial data are not well described by Gaussian processes [9,10,24–30]. However, this study attempts to describe the behaviour of the expected return and the variance of a trading strategy when the price dynamics of an asset is governed by known stochastic processes. This approach paints a picture of how the major system variables relate to each other. More realistic and complex behaviour can be addressed by considering alternative processes for the underlying price dynamics and this is the focus of future study.

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